

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 39

WORLD CLIMATIC TYPES

THE STEPPE CLIMATE

- **ECONOMIC DEVELOPMENT OF PEOPLE**

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT

EXTENSIVE MECHANIZED WHEAT CULTIVATION

TYPE- 2



Mechanized wheat cultivation

1 The temperate grasslands are ideal for **extensive wheat cultivation**.

2 The **cool, moist spring** stimulates **early growth** and the light showers in the ripening period help to swell the grains to ensure a good yield.

3 The **warm, sunny summer** is not only advantageous for harvesting, but also enables the **straw to be dried** for farm use.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

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TYPE- 2



Levelness of the Steppes

4

In addition, the levelness of the Steppes and other temperate grasslands all over the world makes ploughing and harvesting a comparatively easy job.

5

Mechanical ploughs loosen the soil and get the fields ready for sowing in the shortest possible time.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT

EXTENSIVE MECHANIZED WHEAT CULTIVATION

TYPE- 2



6

In the Prairies, the Argentinian Pampas, the Ukrainian Steppes and the Downs of the Australia combine-harvesters, reap, thresh winnow and sack the grains almost as soon as the stalks are gathered.

7

One distinct drawback of this form of extensive mechanized farming is the consequent low yield.

Yield per unit of land is low in extensive mechanized agriculture! THINK WHY?

EXTENSIVE vs INTENSIVE CULTIVATION

1. Extensive agriculture, in agricultural economics, is a system of crop cultivation using small amounts of labour and capital in relation to area of land being farmed.
2. The crop yield in extensive agriculture depends primarily on the natural fertility of the soil, the terrain, the climate, and the availability of water.
3. Extensive agriculture is distinguished from intensive agriculture in that the latter, employing large amounts of labour and capital, enables one to apply fertilizers, insecticides, fungicides, and herbicides and to plant, cultivate, and often harvest mechanically.
4. Because extensive agriculture produces a lower yield per unit of land, its use commercially requires large quantities of land in order to be profitable.
5. This demand for land means that extensive agriculture must be carried on where land values are low in relation to labour and capital, which in turn means that extensive agriculture is practiced where population densities are low and land tracts are available in plenty.

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8 In comparison, the wheat yield in countries that practice intensive farming are much higher, at times almost thrice the yield.

9 This is attributed to the greater attention given to a smaller piece of land, which is not practicable in the extensive wheat-lands where a farmer owns anything from 6000 to 40,000 acres as in the Prairies.

10 But, if we consider the yield per man, this is very much higher in the extensive farms.

11 In this respect, the sparsely populated temperate grasslands of the mid-latitudes produce the greatest quantity of wheat per capita amongst the world's wheat-growing nations.

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THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

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TYPE- 2

12 They are naturally the greatest wheat exporters.

13 Three-quarters of Canada's 10 million tons of annual wheat production is exported, mainly to Europe which does not produce sufficient wheat to feed her very dense population, despite her high wheat yield.

14 Her wheat needs are so great that shipments of wheat and flour arrive at her ports from almost every part of the temperate grasslands, from the USA, Argentina, Uruguay, Australia and the U.S.S.R.

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THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT

EXTENSIVE MECHANIZED WHEAT CULTIVATION

TYPE- 2



15 Three-quarters of the world's wheat is winter wheat, i.e wheat sown in winter.

16 It is a hard wheat with low moisture content and it is best for bread-making and is extensively traded.

17 Poleward, where the winter temperatures are too cold for the wheat seedlings to survive, spring wheat is grown.

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THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT

EXTENSIVE MECHANIZED WHEAT CULTIVATION

TYPE- 2



18 It is the **less important soft wheat**, more suitable for making **cakes, biscuits and pastas** rather than bread.

19 In North America, **winter wheat** is dominant south of the Great Lakes in U.S.A, while **spring wheat** is sown mainly in the **Canadian Prairie** provinces of Alberta, Saskatchewan and Manitoba.

20 In the **warmer, wetter regions**, **maize** is increasingly grown.

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THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT

PASTORAL FARMING

TYPE- 3



1 When pioneer settlers first moved into the temperate grasslands, there were very few animals.

2 The natural conditions suit animal farming.

3 Subsequently, cattle, sheep, pigs and horses were introduced and they proved very successful.

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THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT

PASTORAL FARMING

TYPE- 3



Refrigerated Containers

4

With the development of refrigerated ships in the late nineteenth century, the temperate grasslands became major pastoral regions, exporting large quantities of beef, mutton, wool, hides.

5

Milk, butter, cheese and other dairy products are also important in some parts of the North American grasslands.

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THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT

PASTORAL FARMING

TYPE- 3

6 The development was particularly spectacular in the southern hemisphere, where the winters are milder and the rainfall is more evenly distributed.

7 The original tuft-grass was ploughed up and replaced by the sown alfalfa.

8 The semi-wild cattle in the Pampas were either crossed with or replaced by the imported pedigree stock from Europe.

9 Soon the Pampas became so involved with the pastoral industry that it took the lead in the world's export of beef.

10 The growth was rapid and town like Buenos Aires, Bahia Blanca, Fray Bentos and Montevideo became known throughout the world.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

ECONOMIC DEVELOPMENT

PASTORAL FARMING

TYPE- 3



Merino sheep

11 This is also true of other temperate grasslands. Much beef is produced in the Great Plains of U.S.A and Australia became the world's leading wool exporter, accounting for a third of its production.

12 In the Eurasian Steppes, too, increasing emphasis is being placed on the ranching animals for meat production.

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**THE TEMPERATE CONTINENTAL
(STEPPE) CLIMATE****ECONOMIC DEVELOPMENT**

To sum it up, various grassland regions are famous for different kinds of economic activities. They include:

- Prairies – Wheat cultivation, livestock ranching.*
- Velds – sheep and cattle rearing, maize cultivation*
- Pustaz – wheat and beet sugar cultivation.*
- Pampas – wheat cultivation, export of dairy, beef products.*
- Downs and Canterbury – Wool production from Merino sheep, dairy products*

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao
K. Srikanth
D. Sunil Kumar
L. V. Krishna
Y. Deepthi

Amrita Naidu
M. Binukrishna
G. Ravi Babu
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